

Direct Merging and Contracting Returns for Affine Transfer

Collatz proof project — September 5, 2026

Let T be the shortcut Collatz map, $j_t(n)$ its odd-step count, and $\mathcal{R}(n)$ the assertion that n reaches one. Put $\mathcal{A}(u) := [\mathcal{R}(3u + 2) \Rightarrow \mathcal{R}(27u + 20)]$. The previous formalization proves that $\forall u \geq 0, \mathcal{A}(u)$ is equivalent to full Collatz. Here we give direct and inductive certificates on particular arithmetic progressions; the universal statement remains open.

Theorem 1 (Direct merging). *For every natural Q , $\mathcal{A}(36 + 64Q)$ holds.*

Proof. At the base parameter, $T^6(110) = T^6(992) = 47$, with odd counts three and one. Exact binary-cylinder transport therefore gives

$$T^6(110 + 192Q) = T^6(992 + 1728Q) = 47 + 81Q.$$

Reaching one transfers through the common endpoint in either direction. Lean checks both the base computations and the extension to every Q . $\square$

Theorem 2 (Contracting parameter return). *For every natural Q , set $u = 2308 + 4096Q$ and $v = 45 + 81Q$. Then*

$$v < u, \quad T^{12}(3u + 2) = 3v + 2, \quad T^{12}(27u + 20) = 27v + 20.$$

Consequently $\mathcal{A}(v) \Rightarrow \mathcal{A}(u)$. The premise $\mathcal{A}(v)$ is retained; the theorem does not prove transfer unconditionally on this progression.

Proof. The base endpoints are $T^{12}(6926) = 137$ and $T^{12}(62336) = 1235$, both with four odd steps. Parameter increments of $4096Q$ give endpoint increments $243Q$ and $2187Q$, respectively. These are precisely the increments obtained by replacing the parameter 45 with $45 + 81Q$. Also $u - v = 2263 + 4015Q > 0$. If $3u + 2$ reaches one, so does its shifted value $3v + 2$. Apply $\mathcal{A}(v)$, then transport convergence back from $27v + 20$ to $27u + 20$. Every step is kernel checked in `Collatz.AffinePairReturn`. $\square$

Exact search criterion. On the cylinder $u = r + 2^tQ$, the two endpoints are

$$x + 3^{j+1}Q, \quad y + 3^{k+3}Q,$$

where $x = T^t(3r + 2)$, $y = T^t(27r + 20)$ and j, k are the respective odd counts. Direct merging requires $x = y$ and $j = k + 2$. A return requires $y = 9x + 2$, $j = k$ and $x \equiv 2 \pmod{3}$. Then $v = (x - 2)/3 + 3^jQ$; checking $(x - 2)/3 < r$ and $3^j \leq 2^t$ proves $v < u$ for every quotient. Base endpoint coincidence alone is insufficient for either certificate.

Finite evidence. The reproducible search stops extending a cylinder once either certificate is found. Through depth 18, its 3,236 first certificates classify 30,176 of the 2^{18} parameter residues by direct merging and 1,267 by a contracting return; 230,701 remain unresolved. The return count is conditional induction coverage, not a count of independently proved transfer statements. These census figures are Python evidence, not a kernel completeness theorem. Tests compare the full small-depth census against direct orbits and replay every saved certificate at $Q = 0, 1, 7$. The formulas above, rather than those sample quotients, justify lifting.

A boundary for aligned merging. Lean also proves that both 2 and 20 reach one but $T^t(2) \neq T^t(20)$ for every t . Their joint orbit enters the two-state cycle $(1, 2), (2, 1)$. Thus parameter zero needs a base case or another argument, even though its transfer is true. More generally, an identity on an infinite parameter progression using fixed path lengths s, t forces equality of endpoint slopes. For these partners that equality is $3^{j+1}/2^s = 3^{k+3}/2^t$, hence $s = t$ and $j = k + 2$. The slope deduction is a written arithmetic observation, not an additional Lean theorem. The single exceptional parameter does not refute a proof combining base cases and contracting returns.

Remaining obligation and verification. A complete induction needs every parameter to be a base case, admit a direct certificate, or return to a smaller parameter. No such coverage theorem is proved. Six public declarations in `AffinePairReturn` enter the axiom audit. The search and saved data are `analysis/affine_pair_search.py` and its matching JSON file. This work supplies induction steps, not the full induction, and makes no mathematical priority claim.